MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)
MG3DSEMAT200: ACTURIAL MATHEMATICS

Duration: 2 hrs Maximum Marks: 70

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Answer any five questions. Each question carries 2 marks

1. Explain "time until terminates" random variable. [U][1]
2. Define stochastic process. [K][1]
3. Define Poisson process. [K][2]
4. Define Brownian motion process. [K][3]
5. Define the force of transition from state i to state j at time t. [K][4]
6. Let i be a state such that $\mu(t) = 0$ for all $j \neq i$ and all t. Then prove that for $s < t$ $p_{ii}(s,t) = e^{\int_s^t \mu_{ii}(r)dr}$, and $p_{ji}(s,t) = 0$ for $j \neq i$. [An][4]
7. When do we say that a trading strategy is self financing? [An][5]
8. State the fundamental theorem of asset pricing. Explain with example. [K][5]

Part B
Answer any five questions. Each question carries 6 marks

9. Define Markov chain. Explain with an example. [K, A][1]
10. Define transition matrix. A box contains two balls, either of which can be red or yellow. At each stage a ball is chosen at random and replaced with a ball of the opposite colour. Let X_n be the number of red balls in the box at time n. Find the transition matrix. [K, A][1]
11. Prove that a counting process that has stationary and independent increments is a Poisson process with rateλ if and only if the following hold
(i) $P(N(h) = 1) = \lambda h + o(h)$.
(ii) $P(N(h) \geq 2) = o(h)$ [A][2]

12. Explain geometric Brownian motion . [An][3]

13. Prove that $\sum_{j=0}^{N} \mu_{ij}(t) = 0$ for all i and t. [A][4]

14. For a two state model we have forces of transition given by

$$\mu_{01}(0,t) = \begin{cases} 1, \ if 0 \leq t < 2 \\ 2, \ if 2 \leq < 3. \end{cases}$$

$$\mu_{10}(0,t) = \begin{cases} 2, \ if 0 \leq t < 2 \\ 3, \ if 2 \leq < 3. \end{cases}$$

Find the probability that the process will be in state 1 at time 2.5 given that it is at state 0 at time 0. [An][4]

15. Prove that in an arbitrage free financial market, if there is a self-financing trading stategy for which $V(r)$ is a constant c for some n, then $C = V(0)(1+r)^n$ [A][5]

16. Let X bve the contingent claim that takes the value 60 on UU and 0 elsewhere. Take $r = 0$ Find a self-financing trading strategy to replicate this claim. [C][5]

Part C
Answer any three questions. Each question carries 10 marks

17. Prove that in a finite state stationary Markov chain there is atleast one recurrent state. Also prove that starting from any transient state, we must eventually reach a recurrent state. [A][1]

18. Define rechable state. Prove that if i is recurrent and j is reachable from i then j is recurrent. [A][1]

19. (a) Define nonhomogeneous Poisson process. [K][3]
    (b) What is mean by hitting times? Compute the distribution of $T_a = inf\{t : T(t) = a\}$. [An][3]

20. Explain Brownian motion with drift and geometric Brownian motion. [An][3]

21. (a) Calculate the probability that at time n, the first type of failure will have occured but not the second. [A][4]
    (b) Find the probability that the second cause of failure occures before time n and after the first cause of failure. [E][4]

22. State and prove the fundamental theorem of asset pricing. [An][5].